



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Business Systems

Academic Year: 2024- 2025 (Odd Semester)

Degree, Semester & Branch: B.Tech ,III & CSBS

Course Code & Title: AD3491 & Fundamentals of Data Science and Analytics

Name of the Faculty member: M. Jeya Sundari, AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit V / Linear Least Square
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Flipped classroom

- **Justification:**

Linear least squares are a mathematical approach used to find the best-fit line for a set of data points by minimizing the sum of the squared differences between observed and predicted values. The flipped classroom approach reverses traditional teaching by delivering instructional content outside the classroom (via videos, readings, or online resources) and dedicating in-class time to active learning, problem-solving, and collaborative exercises.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:**

- Understand the concept of Linear Least Squares and its mathematical derivation.
- Learn the significance of minimizing residuals to find the best-fit line.
- Explore real-world applications, such as regression analysis.
- • Explain the concept of Linear Least Squares, including:
 - Importance of minimizing squared residuals.
 - Formula derivation: $y = mx + cy$ using matrices.
- Use visual examples, like fitting a line to scatter plots.
- Review key points from pre-class materials.
- Address misconceptions or difficulties noted in the quiz.
- Provide a small dataset (e.g., hours studied vs. exam scores).

Ask groups to:
- Plot data points.
- Calculate the least squares regression line manually.

➤ Discuss the residuals and overall fit.

➤ **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO1	PSO2
	3	3	2	2	1	2	2	2

• **PSO mapped:**

Practice	PO1	PO2	PO3	PO4	PO9	PSO1	PSO2
	3	3	2	2	2	2	2
Justification for correlation	Needs to learn the basic knowledge of Statics to do the analysis of the problem.	The students will identify the relevant Mathematical Statics concept for a given application.	Interpret the complex Engineering analytic problem With the Help of Linear Square method	The students can identify best Analytic method based on them analysis.	The students can do individually perform the Linear Square analysis	The students can continued upgrading of technical knowledge using the concept of sorting.	The students able to implement different Analytical tools In real time applications

• **Images / Screenshot of the practice:**



- **Reflective Critique:**
- **Feedback of practice from students and other stakeholders:**
 - ✓ Student felt good, since they can study at their own pace/time.
 - ✓ They felt that through such learning, they can explore more.
- **Benefit of the practice:**

Enhanced Learning: Students understand Linear Least Squares through guided exploration and application.

- **Skill Development:** Encourages computational proficiency, critical thinking, and teamwork.
- **Efficient Use of Time:** Theoretical aspects are covered outside class, freeing in-class time for practical work.
- **Student-Centered Learning:** Promotes active engagement and self-paced learning.
- **Challenges faced in implementation:**
 - ✓ Relies on student preparation – few students did not come prepared.
 - ✓ The depth of the subject can be dictated by the student themselves or the group the student is working with.
 - ✓ The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class.

References:

- <https://www.teachthought.com/learning/the-definition-of-theflippedclassroom/#:~:text=A%0flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- https://en.wikipedia.org/wiki/Flipped_classroom
- [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-whatwhy-and-how-to-implement-a-flipped-classroom-model>

Faculty Name / CSBS

HOD / CSBS